Write-up: CHALL

por *dleskow*

Informações do chall

- **Título**: CTF/CHALL
- **Autor**: ??
- **Contexto**: Basta inserir aqui
- **Arquivos**:
  - *chall.zip*
    - *CHALL* (ELF)
    - *libc.so.6*
    - *ld-linux-x86-64.so.2*
    - *Dockerfile*
    - *run.sh*
    - *flag.txt*

1. Write-Up

1.1. Vulnerabilidade

```
┌──(dleskow㉿ganesh)-[/home/ganesh/CTF/CHALL]
└─# Ctrl+C, Ctrl+V terminal commands and/or program execution like this
# Or if you want the xxd of something? idk
┌──(dleskow㉿ganesh)-[/home/ganesh/CTF/CHALL]
└─# xxd file
00000000: 4374 726c 2b43 2c20 4374 726c 2b56 2074 Ctrl+C, Ctrl+V t
00000010: 6572 6d69 6e61 6c20 636f 6d6d 616e 6473 erminal commands
00000020: 206c 696b 6520 7468 6973 0a               like this.
```

```
pwndbg> Ctrl+C, Ctrl+V pwndbg output like this:
pwndbg> info proc m
0x00007f23b8b99000 0x00007f23b8b9b000 0x2000    0x1e6000  rw-p /home/ganesh/CTF/CHALL/libc.so.6

pwndbg> hexdump 0x00007f223e7c0000 0x5000
+16e0 0x7f23b8b9a6e0  08 f0 f0 6c fc 7f 00 00  00 00 00 00 00 00 00 00  |...l....|........|

pwndbg> b* main+208
Breakpoint 2 at 0x562cb46585a1
pwndbg> c
Continuing.

pwndbg> p $rbp
$2 = (void *) 0x7ffc6cf0eef0
```

```
unsigned __int64 __fastcall show_ida_functions(unsigned __int8 a1)
{
  canary = __readfsqword(0x28u);
  printf("Ctrl+C, Ctrl+V IDA functions here: %p\n", offset);
  return canary - __readfsqword(0x28u);
}
```

Ataque utilizando *pwntools*

Ataque

Vazamento de endereços ou sei lá

Descrição

```
def template():
  print('Ctrl+C, Ctrl+V python code like this')
if 2+2 == 5:
  p_equals_np_proof()
else:
  template()
```

Etc

2. solve.py

```
from pwn import *

exe  = ELF("../notes_patched")
libc = ELF("../libc.so.6", checksec=False)
ld   = ELF("../ld-linux-x86-64.so.2", checksec=False)

REMOTE = False
REMOTE = True

if REMOTE:
    context.bits = 64
    p = remote('ganesh.icmc.usp.br', 5005)
    #p = remote('localhost', 5005)
else:
    context.binary = exe
    context.terminal = ['qterminal', '-e']
    p = process([exe.path])
    #gdb.attach(p, exe=exe.path, gdbscript="b* main+413\ncontinue")
    gdb.attach(p, exe=exe.path, gdbscript="continue")

#context.log_level = 'debug'

# Bla
# Bla
# Bla

p.interactive()
```